Timothée Bonnefoi

PhD student at the University of Antwerp

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Current position

December 2022 - present — **PhD student in Mathematics at Universiteit Antwerpen (Belgium) under the supervision of Wendy Lowen and Arne Mertens**,

- Defined a formal notion of sheaf toposes in sufficiently nice 2-categories recovering several characterisations generalising Grothendieck toposes.
- Keywords: Category theory, topos theory, formal category theory, logic, monoidal category, (dependent) type theory.

Research stays

February - July 2022 — **Internship in homotopy type theory**, *University of San Diego (United States)*, Under the supervision of Micheal Shulman

September 2021 - January 2022 — **Internship in topos theory**, *Università degli Studi dell'Insubria (Italy)*, Under the supervision of Olivia Caramello

April - July 2021 — **Remote internship in categorical probability theory**, *FMF, University of Ljubljana (Slovenia)*, Under the supervision of Alex Simpson

May - July 2020 — **Remote internship in general relativity**, *Université de Tours (France)*, Under the supervision of Romain Gicquaud

June - July 2019 — **Internship in rough path theory**, *University of Bath (United Kingdom)*, Under the supervision of Hendrick Weber

June - July 2018 — **Internship in experimental particle physics**, *LAPP Annecy (France)*, under the supervision of Francesco Costanza and Stéphane Jézéquel in the ATLAS group

Education

Spring 2027 (Expected) — **Doctor of Science: Mathematics**, *Universiteit Antwerp*

2022 — **Diploma of the ENS Lyon (Ecole Normale Supérieure Lyon)**,

- *Majored in Mathematics and Physics*, Minored in Computer Science

2021 — **Master Mathématiques avancées (Advanced mathematics)**

2020 — **Master 1 Sciences de la Matière (Materials sciences)**

2018 — **Bachelor of Sciences: Physics**

Publications

- **A priori bounds for rough differential equations with a non-linear damping term** joint work with Ajay Chandra, Augustin Moinat, and Hendrik Weber, *Journal of Differential Equations*, Volume 318, 2022, Pages 58-93, ISSN 0022-0396, ⟨doi|preprint⟩.

Teaching experience

Fall 2024/2025 — **Teaching assistant for the Bachelor 3 class 'Number Theory' at the University of Antwerp taught by Julian Lyczak.**, *Exercise sessions, homework and examination correction*

Spring 2024/2025/2026 — **Teaching assistant for the Bachelor 2 class 'Algebra III: Fields' at the University of Antwerp taught by Julian Lyczak.**, *Exercise sessions, homework and examination correction*

Fall 2023 — **Teaching assistant for the Bachelor 3 class 'Fields and Galois Theory' at the University of Antwerp taught by Karim Johannes Becher.**, *Exercise sessions, homework and examination correction*

Spring 2023 — **Teaching assistant for the Bachelor 2 class 'Algebra III: Fields' at the University of Antwerp taught by Ignacio Barros.**, *Exercise sessions, homework and examination correction*

Talks

September 2026 — **Contributed talk 'Formal topos theory' at PSSSL113 (Louvain-la-Neuve)**

Conferences

June 2024 — **Attended the conference CT24 in Santiago de Compostella**

2022 - current — **Attending presentations of the Antwerp Algebra seminar**

June 2022 — **Attended the HoTT Muri 2022 meeting**

Scholarships

2022-2026 — **Dehousse doctoral scholarship in Mathematics of the University of Antwerp**

Additional activities

2022-2026 — **Doctoral study program of the University of Antwerp**

March 2023 — **Certificate 'Mind the gap' for good academic practices**

Languages

French — Native

English — CERCL Level C1, CAE (Cambridge English Advanced) certificate

Dutch and Spanish — Intermediate reading level, Basic speaking level

Computer science and programming skills

OS	GNU/Linux: ArchLinux, NixOS (administrator) Windows.	Programming languages	Python: Advanced usage.
Tools	LaTeX, Typst, Git		OCaml, SQL, Coq,C/C++, Nix, Emacs Lisp: Basic use

Theory Courses of the Bachelor 3 & Master 1 in Computer Science of ENS Lyon, in particular Algorithms and Programming